

Food Delivery Speed Prediction Using Machine Learning

Faisal Navid
23101121

Shuvo Das
23201558

Abstract

This project focuses on predicting food delivery speed using machine learning techniques. The delivery speed is categorized into three classes: Fast, Average, and Slow. The dataset includes various features such as distance, preparation time, traffic conditions, and weather. Data preprocessing techniques including handling missing values, encoding categorical variables, and feature scaling were applied to prepare the dataset. Multiple supervised learning models, including a Neural Network, Decision Tree, and K-Nearest Neighbors (KNN), were trained and evaluated. Additionally, K-Means clustering was applied as an unsupervised learning approach. The models were evaluated using accuracy, precision, recall, and F1-score. The results demonstrate that machine learning models can effectively predict delivery speed and highlight the importance of proper preprocessing and model selection.

Keywords— Machine Learning, Classification, Neural Network, KNN, Decision Tree, K-Means

I. Introduction

Food delivery systems have become an essential part of modern urban life. Efficient and timely delivery is crucial for customer satisfaction and operational success. This project aims to predict the delivery speed of food orders based on multiple influencing factors.

The delivery speed is classified into three categories: Fast, Average, and Slow. The prediction is based on features such as delivery distance, preparation time, traffic conditions, and weather. Understanding how these factors affect delivery performance can help optimize logistics and improve service quality.

This problem is formulated as a multi-class classification task. Several supervised machine learning models are implemented and compared, along with an unsupervised clustering technique to analyze patterns in the data.

II. Dataset Description

The dataset consists of food delivery records with multiple features influencing delivery speed.

- **Number of Data Points:** [Insert value]
- **Number of Features:** [Insert value]

Feature Types:

- **Numerical Features:** Distance, Preparation Time, Delivery Time
- **Categorical Features:** Traffic Level, Weather Condition, Delivery Speed

The target variable is **Delivery Speed**, which has three classes: Fast, Average, and Slow.

III. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset and extract meaningful insights.

Key Observations:

- Shorter distances are generally associated with faster deliveries.
- High traffic levels significantly slow down delivery speed.
- Longer preparation times contribute to delayed deliveries.
- Some numerical features contain outliers that may impact model performance.
- Correlation analysis shows strong relationships between certain features and the target variable.

EDA helped identify patterns and guided preprocessing and model selection.

IV. Data Preprocessing

Data preprocessing was performed to ensure the dataset is clean and suitable for machine learning models.

A. Handling Missing Values

Missing values were handled using:

- Mean/median for numerical features
- Mode for categorical features

B. Removing Duplicates

Duplicate records were removed to avoid bias.

C. Dropping Irrelevant Features

Features such as Order_ID were removed as they do not contribute to prediction.

D. Encoding Categorical Variables

Categorical variables were converted into numerical form using encoding techniques. Ordinal encoding was applied where appropriate (e.g., traffic levels).

E. Feature Selection

Feature selection techniques were used to identify the most relevant features for prediction.

F. Feature Scaling

Standardization was applied using StandardScaler to normalize feature values, which is particularly important for distance-based models like KNN and Neural Networks.

V. Methodology

A. Neural Network

A Multi-Layer Perceptron (MLP) classifier was used to model complex non-linear relationships in the data. It consists of multiple hidden layers and uses activation functions to learn patterns.

B. Decision Tree

The Decision Tree model splits the dataset into branches based on feature values, forming a tree-like structure of decision rules. It is easy to interpret and visualize.

C. K-Nearest Neighbors (KNN)

KNN is a distance-based algorithm that classifies a data point based on the majority class of its nearest neighbors. Feature scaling is essential for this model.

D. K-Means Clustering

K-Means clustering was applied as an unsupervised learning technique to group the data into clusters. The number of clusters was set to three to match the number of target classes.

VI. Model Evaluation

The models were evaluated using the following metrics:

- **Accuracy:** Overall correctness of predictions
- **Precision:** Correctness of positive predictions
- **Recall:** Ability to identify actual positives
- **F1 Score:** Balance between precision and recall

Confusion Matrix

Confusion matrices were used to visualize model performance by comparing predicted and actual values.

Comparison

The performance of Neural Network, Decision Tree, and KNN models was compared based on the above metrics. Models that achieved higher F1-scores were considered more reliable.

VII. Results and Discussion

The results show that all three models are capable of predicting delivery speed with reasonable accuracy.

- The **Neural Network** performed well in capturing complex relationships.
- The **Decision Tree** provided interpretable decision rules.
- The **KNN model** performed effectively after feature scaling but is sensitive to distance calculations.

The comparison highlights that model performance depends on both data preprocessing and algorithm selection.

VIII. Conclusion

This project demonstrates the application of machine learning techniques to predict food delivery speed. The dataset was analyzed, preprocessed, and used to train multiple models.

Among the models, Neural Network, Decision Tree, and KNN showed effective performance in classification tasks. The results emphasize the importance of proper preprocessing, feature selection, and evaluation.

Future improvements may include hyperparameter tuning, using larger datasets, and applying more advanced machine learning techniques.